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MAMUZA ENGINEERING
TRAINING & OUTREACH DEPARTMENT

FULLY SPONSORED OUTREACH PROGRAM

Smart IoT & Embedded Systems Program

Learner Handbook — an online, project-based introduction to IoT and embedded systems for tertiary students, ending in a connected smart-irrigation capstone build.

DURATION	SESSIONS	FORMAT	COST TO YOU
4 Weeks	8 Live Sessions	Fully Online	FREE

Welcome & Table of Contents

Karibu, and congratulations on securing a place on the **Mamuza Engineering Smart IoT & Embedded Systems Program** — a fully sponsored outreach initiative designed to give tertiary students real, hands-on exposure to the Internet of Things using industry-relevant hardware, from the comfort of your own campus or home.

Over four weeks you will move from your very first blinking LED to a fully connected, cloud-controlled smart-irrigation system that you design, wire, code, and demonstrate live — using a personal component kit that is yours to keep.

This handbook is your companion for the entire program. Keep it open beside you in every session — it contains your session-by-session objectives, the components you'll need, your assignments, and everything you need to know about how the program runs.

In this handbook

01	About the Program & About Mamuza Engineering	3
02	How the Program Works (Online Learning Guide)	4
03	Program Roadmap — 4 Weeks at a Glance	5
04	Your Component Kit	6
05	Week 1 — Foundations of IoT & Embedded C	8
06	Week 2 — Sensing the Environment	10
07	Week 3 — Multi-Sensor Systems & Actuation	11
08	Week 4 — Cloud Connectivity & Capstone Build	12
09	Assessment & Certification	14
10	Learner Code of Conduct & Support	15

About the Program

The **Smart IoT & Embedded Systems Program** is a Mamuza Engineering community outreach initiative created to open the world of embedded electronics and the Internet of Things to tertiary students — no matter their background, campus, or prior experience with hardware.

It is a **fully sponsored, project-based online program**. Every learner who is accepted receives a personal hardware kit at no cost, joins live instructor-led sessions twice a week, and works through a structured build path that ends in a real, working, cloud-connected smart-irrigation system — designed, wired, and coded by you.

Who this is for

- Tertiary (college & university) students, any year of study
- No prior electronics or coding experience required
- Curiosity about hardware, coding, or engineering
- Access to a laptop and stable internet for live sessions

What you'll walk away with

- Practical embedded C/C++ programming skills
- Hands-on experience with real sensors and actuators
- A working IoT capstone project for your portfolio
- A Mamuza Engineering certificate of completion
- Your component kit, yours to keep and build on

Program Snapshot

DETAIL	INFORMATION
Duration	4 weeks
Sessions	8 live sessions — 2 per week, 2 hours each
Format	Fully online (live instructor-led + guided self-practice)
Stack	C/C++ (Embedded), ESP32, Arduino IDE, Blynk IoT Platform
Capstone	Session 8 — Automated Smart Irrigation & Farm System
Cost to learner	Fully sponsored — kit, tuition, and certification at no charge

About Mamuza Engineering

Mamuza Engineering is an engineering and technology training company committed to building practical, future-ready skills in embedded systems, IoT, and automation. Through our Training Department, we run outreach programs that put real hardware into the hands of students and early-career engineers — because the fastest way to learn engineering is to build something that works.

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SECTION02

How the Program Works

Session Format

Each of the 8 sessions runs for 2 hours and is delivered live online. A typical session follows this rhythm:

SEGMENT	APPROX. TIME	WHAT HAPPENS
Concept Walkthrough	25 min	Your instructor introduces the theory behind that session's topic with visuals and live demos.
Guided Build	60 min	You wire your kit and write code alongside the instructor, step by step, with your camera and workspace visible for support.
Troubleshooting & Q&A	20 min	Debugging support, common-mistake fixes, and open questions.
Assignment Briefing	15 min	The instructor explains that week's take-home assignment and deliverable.

What You Need on Your End



Setup instructions for the Arduino IDE, ESP32 board drivers, and your Blynk account will be shared before Session 1 in your onboarding email.

Attendance & Assignments

- Live attendance at all 8 sessions is expected — this is a hands-on, sequential build, and each session builds on the last.
- Every session (except the capstone) closes with a short take-home assignment, due before the next session.
- Assignments are submitted as a short video/photo of your working circuit plus your code — details are shared in each session.
- Sessions are recorded and shared with registered learners for revision only.

Because this is a sponsored kit, please treat your components with care — you keep them permanently at the end of the program, so they're yours to keep experimenting with long after Session 8.

Program Roadmap — 4 Weeks at a Glance

Here is the full arc of the program. Each week has a theme, and each session has a clear, standalone build — but everything you make in Weeks 1–3 becomes a building block in your Week 4 capstone.

WEEK	SESSION	TITLE	CORE FOCUS
1	1	Introduction to IoT & Hardware Setup	IoT architecture, embedded C basics, breadboard power & LEDs
	2	Digital & Analog Inputs	Buttons, touch sensing, potentiometers, PWM & RGB color mixing
2	3	Motion Detection & Alarms	PIR motion sensing, tone-based sirens, security logic
	4	Climate Sensing & OLED Display	DHT11 sensor, I2C protocol, live data on an OLED screen
3	5	Multi-Sensor Farm Hub	Soil moisture + rain sensing, combined environmental dashboard
	6	Relay Driving & Load Safety	Switching a water pump safely with a relay & watchdog timing
4	7	Cloud Connectivity with Blynk	Wi-Fi telemetry, virtual pins, live mobile dashboards
	8	Capstone: Smart Irrigation System	Full-system integration & live demonstration

The Build You're Working Towards

By Session 8, every skill from the previous seven sessions comes together into one connected system: an **autonomous smart-farm unit** that reads temperature, humidity, soil moisture and rain, displays it all locally on an OLED screen, automatically triggers a water pump when the soil is dry, sounds alerts when conditions demand attention, and streams all of it to a live dashboard on your phone — with the ability to override the pump remotely.

WHY THIS MATTERS

This is the same underlying architecture used in real precision-agriculture systems and autonomous greenhouses: sensors feeding a microcontroller, which drives local displays and alerts, actuates hardware, and reports to the cloud for remote monitoring and control.

SECTION04

Your Component Kit

Every learner receives the full kit below, shipped to you before Session 1. This is your kit to keep — treat it as the start of your own personal electronics toolbox.

ESP32 Development Board	x1	0.96" OLED Display (I2C)	x1
Single-Color LEDs	x3	RGB LED	x1
DHT11 Temperature & Humidity Sensor	x1	Soil Moisture Sensor Module	x1
Rain Sensor Module	x1	PIR Motion Sensor	x1
Capacitive Touch Sensor Module	x1	Submersible Water Pump	x1
1-Channel 5V Relay Module	x1	Active Buzzer	x1
Passive Buzzer	x1	Potentiometer (10kΩ)	x1
Push Buttons	x3	330Ω Resistors	x3
Breadboard Power Supply Module	x1	6× AAA Battery Pack	x1
Solderless Breadboard	x1	Jumper Wires (M–M, M–F, F–F)	

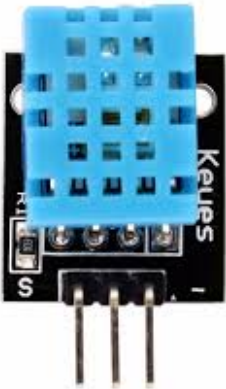
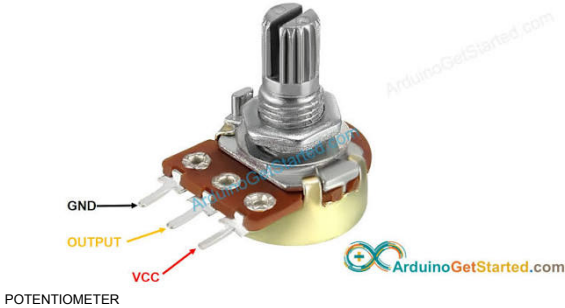
Unbox & check: when your kit arrives, lay every item out against this list before Session 1. Missing or damaged components should be reported to info@mamuzaengineering.com right away so a replacement can reach you in time.

Kit Component Gallery

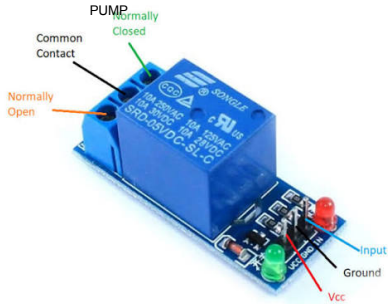
Reference photos of each component are added below — use this page to match what's in the box to what's on your bench.



ESP32



DHT11



5VRELAY



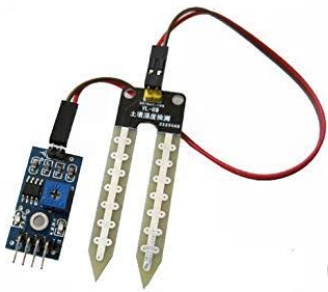
LED



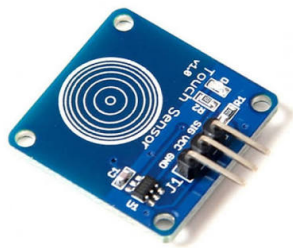
BREADBOARD



POWER MODULE



SECTION04—CONTINUED



TOUCH SENSOR



OLED



PIR MOTION SENSOR

SECTION05—WEEK 1

Foundations of IoT & Embedded C

WEEK 1 · SESSIONS 1 – 2

1

Introduction to IoT, Embedded Systems & Hardware Setup

SESSION 1 · WEEK 1

What You'll Learn

- The IoT architecture: Sensors → Microcontroller → Connectivity → Cloud → User App
- The embedded C flow: header includes, `setup()`, and the infinite `loop()`
- Breadboard power rails, common ground, and current-limiting for LEDs

Session Objectives

- Set up breadboard power rails from the power module and battery pack
- Establish a shared ground between the ESP32 and the external rail
- Write and upload C++ that sequences 3 LEDs with correct timing via functions

Components You'll Use

- ESP32 Dev Board
- Breadboard + Power Module
- 6× AAA Battery Pack
- LEDs ×3, 330Ω Resistors ×3
- Jumper wires

REAL-LIFE APPLICATION

Automated traffic signals rely on this exact idea — microcontroller timing loops driving lights and relays without human intervention.

SECTION 04 — CONTINUED

ASSIGNMENT 1

Modify your `cycleLights()` function so the Red LED blinks 3 times (500ms on/off) as a warning before switching to Green.

Deliverable: Your sketch (code) + a 15-second video of the sequence running.

2

Digital & Analog Inputs — Buttons, Touch & PWM Color Control

SESSION 2 · WEEK 1

What You'll Learn

- Digital HIGH/LOW reads and floating-pin protection via `INPUT_PULLUP`
- Capacitive touch sensing fundamentals
- `if/else` and `switch/case` logic; the ESP32's 12-bit ADC (0–4095)
- PWM theory and using `map()` to mix RGB colors

Session Objectives

- Wire push buttons and a touch module without shorting logic pins
- Read a potentiometer and map it to an 8-bit PWM range (0–255)
- Combine inputs into a multi-state LED, buzzer & RGB interface

Components You'll Use

- Push Buttons ×2, Touch Sensor Module
- 10kΩ Potentiometer
- RGB LED, LED + 330Ω Resistor(s)
- Active Buzzer
- Breadboard + Jumpers

REAL-LIFE APPLICATION

Industrial dual-palm safety switches and smart bulbs both use these exact principles — combined-input logic for safety, and PWM for smooth color and brightness control.

ASSIGNMENT 2

Build a **Safety Lock-Out**: the LED turns ON only when Button 1 AND the touch sensor are triggered together (releasing either turns it off and sounds a short beep). Then add a second button that "locks" the current RGB color mix from the potentiometer until pressed again.

Deliverable: C++ snippet + a short circuit demo video.

Sensing the Environment

WEEK 2 · SESSIONS 3 – 4

3

Motion Detection & Frequency-Based Alarms

SESSION 3 · WEEK 2

What You'll Learn

- PIR sensing — how it detects change in infrared/thermal radiation
- Frequency-based tone generation vs. simple on/off buzzing
- Using `tone()` / `ledcWriteTone()` to build multi-pitch siren patterns

Session Objectives

- Wire and calibrate a PIR sensor's warm-up delay
- Generate a multi-tone siren on a passive buzzer
- Build a security alarm that reacts to motion in real time

Components You'll Use

- PIR Motion Sensor
- Passive Buzzer
- RGB LED + 330Ω Resistors
- Breadboard + Jumpers

REAL-LIFE APPLICATION

Commercial intrusion-detection grids pair PIR sensors with sounders and strobes — the same building blocks you're wiring here.

ASSIGNMENT 3

Wire the touch sensor as a **Disarm Pad**: touching it during an alarm silences the buzzer and holds the RGB LED solid green for 10 seconds before re-arming.

Deliverable: Schematic + working C++.

Climate Sensing & OLED Screen Integration

SESSION 4 · WEEK 2

What You'll Learn

- Single-wire protocol parsing on the DHT11 (temperature + humidity)
- I2C basics: SDA/SCL, device addressing (e.g. 0x3C), bus speed
- Framebuffer text/shape rendering via Adafruit_SSD1306

Session Objectives

- Read temperature (°C) and humidity (%) from the DHT11
- Scan and confirm I2C addresses with a scanner sketch
- Print live sensor metrics to the OLED with clean typography

Components You'll Use

- DHT11 Sensor
- 0.96" I2C OLED
- Breadboard + Jumpers

REAL-LIFE APPLICATION

Smart thermostats sample room air continuously and show live diagnostics locally, exactly as you'll do here.

ASSIGNMENT 4

Wire Button 1 to toggle the OLED reading between °C and °F without interrupting sampling.

Deliverable: Commented C++ including the $F = C \times 1.8 + 32$ conversion.

Multi-Sensor Systems & Actuation

WEEK 3 · SESSIONS 5 – 6

5

Multi-Sensor Farm Environment Hub

SESSION 5 · WEEK 3

What You'll Learn

- Resistive analog sensing: soil moisture and rain-plate conductivity
- Calibrating raw analog readings into dry/wet thresholds in code
- Combining multiple sensor streams into one alerting hub

Session Objectives

- Wire the rain and soil-moisture sensors alongside the DHT11 and OLED
- Convert raw analog readings into 0–100% scales
- Render all farm metrics on-screen and trigger threshold alerts

Components You'll Use

- DHT11 Sensor, Soil Moisture Sensor, Rain Sensor
- 0.96" OLED
- Passive Buzzer + RGB LED
- Breadboard + Jumpers

REAL-LIFE APPLICATION

Precision-agriculture field stations combine soil, rain, and temperature data exactly like this to inform irrigation decisions.

ASSIGNMENT 5

Drive the RGB LED as a health indicator — solid green (nominal), blinking yellow (soil <30%), blinking red + tone (rain detected).

Deliverable: Integrated C++ handling all three sensor states.

6

Relay Driving & High-Power Load Safety

SESSION 6 · WEEK 3

What You'll Learn

- Load isolation with a 5V electromechanical relay (NO/NC/COM)
- Protecting logic pins from back-EMF via optocoupler isolation
- Routing external battery power separately from logic power

Session Objectives

- Wire a 5V relay to switch a submersible pump on external battery power
- Share ground correctly while keeping high-current traces separate
- Write non-blocking C++ pump cycles driven by soil moisture

Components You'll Use

- Soil Moisture Sensor
- 1-Channel 5V Relay
- Submersible Water Pump
- Power Module + 6× AAA Pack

REAL-LIFE APPLICATION

Industrial irrigation systems use relays to let low-voltage controllers safely switch high-power pumps and solenoids.

ASSIGNMENT 6

Add a watchdog cut-off: the pump runs for a maximum of 5 seconds continuous, then forces a 10-second cool-down regardless of moisture readings.

Deliverable: C++ using non-blocking `millis()` timing.

Cloud Connectivity & Capstone Build

WEEK 4 · SESSIONS 7 – 8

7

IoT Cloud Control with the Blynk Platform

SESSION 7 · WEEK 4

What You'll Learn

- Cloud IoT architecture: connecting edge devices over Wi-Fi
- Telemetry and remote control via virtual pins (V0, V1...)
- Building a live mobile/web dashboard for monitoring and override

Session Objectives

- Provision Wi-Fi credentials and link the ESP32 to Blynk Cloud
- Configure datastreams for soil-moisture telemetry and pump control
- Build a dashboard showing live data with remote pump override

Components You'll Use

- Soil Moisture Sensor
- 5V Relay + Pump
- Power Module + 6× AAA Pack
- Wi-Fi (hotspot or router)

REAL-LIFE APPLICATION

Commercial growers monitor soil condition and trigger irrigation remotely via smartphone apps — you're building the same pipeline.

ASSIGNMENT 7

Add a Mode Select switch (V2): **Automatic** (pump follows soil moisture) vs. **Manual** (pump follows app toggle only).

Deliverable: Screen recording of the working Blynk dashboard controlling your hardware.

8

Final Capstone Build — Automated Smart Irrigation & Farm System

SESSION 8 · WEEK 4

What You'll Learn

- Full-system integration: inputs, outputs, and cloud telemetry in one architecture
- Non-blocking design with `millis()` so networking, display, and sensing run smoothly together
- Field-deployment concerns: power stability, Wi-Fi reconnection, housing

Session Objectives

- Assemble the complete system on one breadboard
- Flash unified firmware: sensing, display, automated actuation, and Blynk telemetry together
- Present a working automated smart-farm system to the class

Components You'll Use

- ESP32, 0.96" OLED, DHT11
- Soil Moisture & Rain Sensors
- 5V Relay + Pump
- Passive Buzzer + RGB LED
- Power Module + 6× AAA Pack, full jumper kit

System Assembly Workflow

1. **Power rail setup** — mount the ESP32 and power module on opposite ends of the breadboard, connect the AAA pack at 5V output, and tie all grounds together.
2. **Hardware wiring** — wire the OLED to I2C (SDA/SCL), connect the DHT11, soil moisture and rain sensors, wire the buzzer and RGB LED with resistors, and route pump power through the relay's NO terminal.
3. **Blynk dashboard provisioning** — Soil Moisture % (V0, gauge), Temperature & Humidity (V1/V2, displays), Auto/Manual Mode (V3, toggle), Manual Pump Override (V4, button).
4. **Firmware & testing** — render live climate/soil data on the OLED; when soil <30% and Auto Mode is on, trigger the relay, sound a brief alert, and push the update to Blynk; allow manual override from the app at any time.

REAL-LIFE APPLICATION

Autonomous greenhouse systems manage climate, hydration, local display, alarms, and remote cloud control in exactly this kind of connected package.

Assessment & Certification

Your progress is assessed in two ways: weekly assignments (Sessions 1–7) and your final capstone build (Session 8). Completing the program in good standing earns you a **Mamuza Engineering Certificate of Completion**.

Final Capstone Assessment Rubric (Session 8)

CRITERION	WEIGHT	WHAT'S ASSESSED
Hardware Assembly & Power Management	25%	Clean wiring, correct current-limiting resistors, stable common ground.
Local Display & Sensing	25%	Accurate real-time climate/soil data, cleanly rendered on the OLED.
Automated Actuation & Safety	25%	Reliable pump activation in dry conditions with working timing safety limits.
Cloud Telemetry & Control	25%	Live data streaming and reliable remote override via Blynk.

Certification Requirements

- Attend at least 7 of the 8 live sessions
- Submit at least 6 of the 7 weekly assignments
- Complete and demonstrate the Session 8 capstone build

Certificates are issued digitally within two weeks of the program's end and are shareable on LinkedIn and other professional platforms.

Learner Code of Conduct & Support

What We Ask of You

- Join sessions on time, with your kit, laptop, and workspace ready
- Keep your camera on during the guided-build segment where possible, so instructors can support you
- Be patient and respectful with fellow learners and instructors — everyone starts as a beginner
- Do your own wiring and coding — ask for help, don't copy answers
- Communicate early if you'll miss a session or need more time on an assignment

Getting Help

If a component isn't behaving, your code won't compile, or you're simply stuck — reach out. Support is part of the program, not an afterthought.

NEED	CONTACT
Technical / assignment support	Your session instructor, live in class or on the program group chat
Kit issues (missing/damaged parts)	info@mamuzaengineering.com
General program questions	info@mamuzaengineering.com / 0745 927 033

Welcome aboard. You're joining a community of students who are learning to build the systems shaping modern agriculture, industry, and everyday life. Come curious, get your hands dirty with the wiring, and don't be afraid to break things — that's how engineers learn.



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